

Labsheet-3

Name: Ritik Saini

CU-ID:CU24250087

Roll No.:50

```
import pandas as pd

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_and_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-selectedcolumns.csv")

print(df.head())
print(df.shape)
print(df.columns.tolist())
```

	Age	Attrition	BusinessTravel	DailyRate	Department
0	41	Yes	Travel_Rarely	1102	Sales
1	49	No	Travel_Frequently	279	Research & Development
2	37	Yes	Travel_Rarely	1373	Research & Development
3	33	No	Travel_Frequently	1392	Research & Development
4	27	No	Travel_Rarely	591	Research & Development

EmployeeNumber	DistanceFromHome	Education	EducationField	EmployeeCount
0	1	2	Life Sciences	1
1				
1	8	1	Life Sciences	1
2				
2	2	2	Other	1
4				
3	3	4	Life Sciences	1
5				
4	2	1	Medical	1
7				

```
(1470, 10)
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
'EmployeeNumber']
```

```
import numpy as np import
pandas as pd import
matplotlib.pyplot as plt from
scipy import stats import
statsmodels.api as sm

print("Mean:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mean())
```

```

print("\nMedian:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].median())

print("\nMode:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mode().iloc[0])

print("\nVariance:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].var())

print("\nStandard Deviation:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].std())

```

```

Mean:
Age          36.923810
DailyRate    802.485714
DistanceFromHome  9.192517
dtype: float64

```

```

Median:
Age          36.0
DailyRate    802.0
DistanceFromHome  7.0
dtype: float64

```

```

Mode:
Age          35
DailyRate    691
DistanceFromHome  2
Name: 0, dtype: int64

```

```

Variance:
Age          83.455049
DailyRate    162819.593737
DistanceFromHome  65.721251
dtype: float64

```

```

Standard Deviation:
Age          9.135373
DailyRate    403.509100
DistanceFromHome  8.106864
dtype: float64

```

```

print(df[["Age", "DailyRate", "DistanceFromHome"]].describe())

```

	Age	DailyRate	DistanceFromHome
count	1470.000000	1470.000000	1470.000000
mean	36.923810	802.485714	9.192517
std	9.135373	403.509100	8.106864

```

min      18.000000    102.000000          1.000000
25%     30.000000    465.000000          2.000000
50%     36.000000    802.000000          7.000000
75%     43.000000   1157.000000         14.000000
max      60.000000   1499.000000         29.000000
numeric_df = df[["Age", "DailyRate", "DistanceFromHome"]]
correlation = numeric_df.corr() print(correlation)

```

	Age	DailyRate	DistanceFromHome
Age	1.000000	0.010661	-0.001686
DailyRate	0.010661	1.000000	-0.004985
DistanceFromHome	-0.001686	-0.004985	1.000000

```

plt.figure(figsize=(8, 6))

plt.imshow(correlation, cmap="coolwarm")
plt.colorbar()

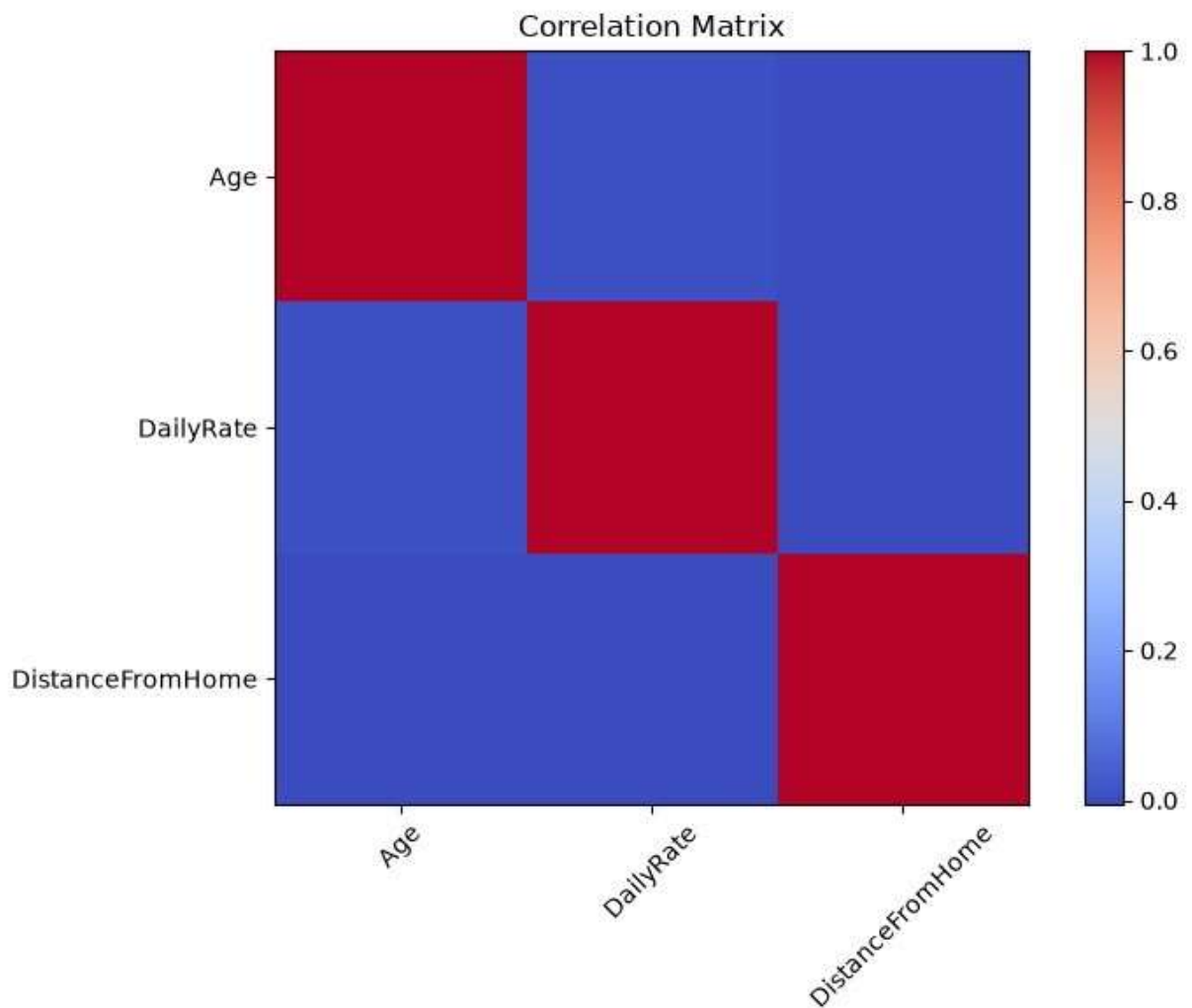
plt.xticks(
    range(len(correlation.columns)),
    correlation.columns, rotation=45
)

plt.yticks(
    range(len(correlation.columns)),
    correlation.columns )

plt.title("Correlation Matrix")

plt.tight_layout()
plt.show()

```



```
r, p = stats.pearsonr(df["Age"], df["DailyRate"])
print("Pearson Correlation:", r) print("P-
value:", p)
Pearson Correlation: 0.01066094264553816
P-value: 0.6829722453880914

left = df[df["Attrition"] == "Yes"]["Age"]
stayed = df[df["Attrition"] == "No"]["Age"]

print("Employees who left:", len(left))
print("Employees who stayed:", len(stayed))
Employees who left: 237
Employees who stayed: 1233
```

```

t_stat, p_value = stats.ttest_ind(
left,      stayed,
    equal_var=False
)

print("T-statistic:", t_stat) print("P-
value:", p_value)

T-statistic: -5.82801185398895
P-value: 1.3797600649439802e-08

if p_value < 0.05:
    print("Reject H0")
    print("There is a significant difference in Age between employees
who left and stayed.") else:    print("Fail to reject H0")
    print("There is no significant difference in Age between employees
who left and stayed.")

Reject H0
There is a significant difference in Age between employees who left
and stayed. print(df["Department"].unique())

<StringArray>
['Sales', 'Research & Development', 'Human Resources']
Length: 3, dtype: str

groups = []

for department in df["Department"].unique():
    groups.append(
        df[df["Department"] == department]["Age"]
    )

f_stat, p_value_anova = stats.f_oneway(*groups)

print("F-statistic:", f_stat) print("P-
value:", p_value_anova)
F-statistic: 0.7655031924976907
P-value: 0.4652855299935297

if p_value_anova < 0.05:
print("Reject H0")
    print("There is a significant difference in Age among
Departments.") else:    print("Fail to reject H0")

```

```
print("There is no significant difference in Age among  
Departments.")
```

Fail to reject H0 There is no significant difference in Age
among Departments.

```
X = df["Age"] y =  
df["DailyRate"] X =  
sm.add_constant(X) model =  
sm.OLS(y, X).fit()  
print(model.summary())
```

OLS Regression Results

```
=====
```

Dep. Variable:	DailyRate	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.001
Method:	Least Squares	F-statistic:	0.1669
Date:	Mon, 31 Aug 2026	Prob (F-statistic):	0.683
Time:	08:31:36	Log-Likelihood:	-10906.
No. Observations:	1470	AIC:	2.182e+04
Df Residuals:	1468	BIC:	2.183e+04
Df Model:	1		

Covariance Type: nonrobust

```
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	785.0985	43.847	17.905	0.000	699.089	871.108
Age	0.4709	1.153	0.408	0.683	-1.790	2.732

```
=====
```

Omnibus:	1174.453	Durbin-Watson:	
----------	----------	----------------	--

```
=====
```

2.082

```
Prob(Omnibus) :          0.000   Jarque-Bera  (JB) :  
88.696  
Skew:          -0.004   Prob(JB) :  
5.49e-20  
Kurtosis:      1.797   Cond. No. :  
159.
```

```
=====
```

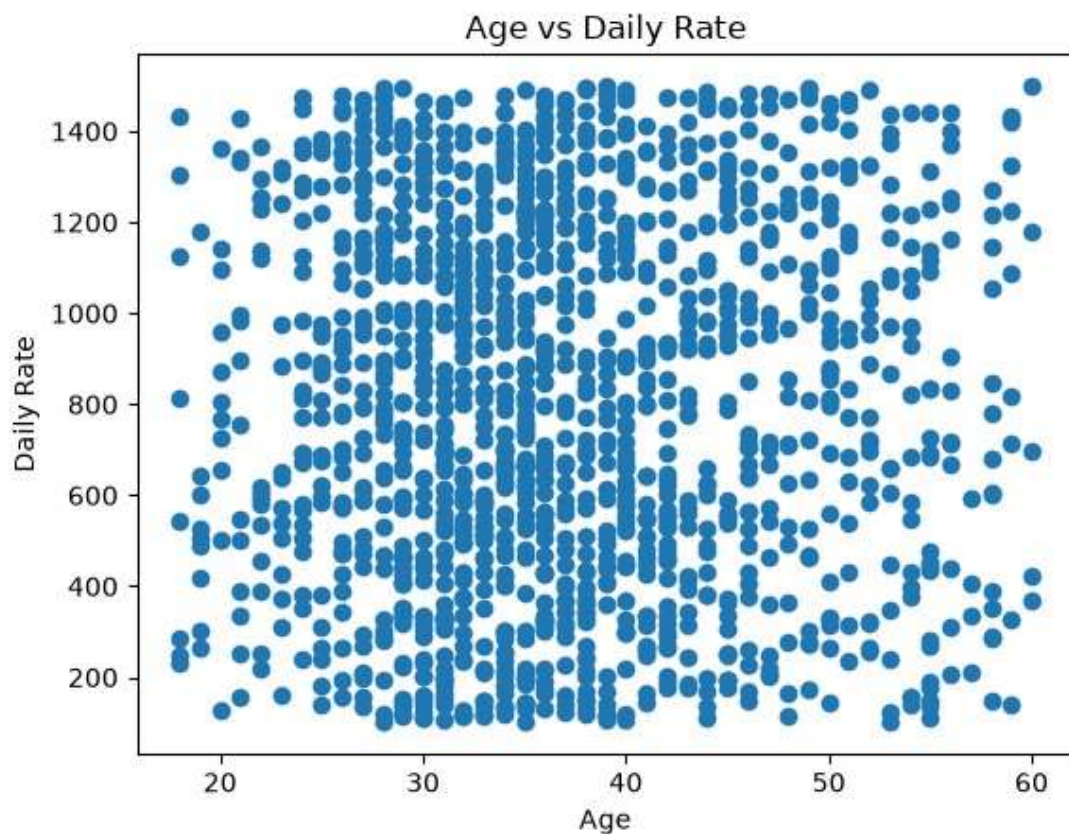
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
plt.scatter(df["Age"], df["DailyRate"])
```

```
plt.xlabel("Age")  
plt.ylabel("Daily Rate")  
plt.title("Age vs Daily Rate")
```

```
plt.show()
```



```
print("R-squared:", model.rsquared)
```

```
print("Regression Coefficients:")
print(model.params)
```

```
Regression Coefficients:
const      785.098535
Age         0.470893
dtype: float64
```

```
print("P-values:")
print(model.pvalues)
```

```
P-values: const
5.267571e-65 Age
6.829722e-01 dtype:
float64
```

```
print("Confidence Intervals:")
print(model.conf_int())
```

```
Confidence Intervals:
              0              1
const  699.089362  871.107708
Age     -1.790341    2.732128
print(model.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          DailyRate    R-squared:
0.000
Model:                  OLS          Adj. R-squared:
-0.001
Method:                 Least Squares    F-statistic:
0.1669
Date:                   Mon, 31 Aug 2026    Prob (F-statistic):
0.683
Time:                   08:31:37          Log-Likelihood:
-10906.
No. Observations:      1470             AIC:
2.182e+04
Df Residuals:          1468             BIC:
2.183e+04
Df Model:               1
Covariance Type:       nonrobust
```

```
=====
=====
```

```
          coef      std err          t      P>|t|      [0.025
0.975]
```

```
-----
-----
```

```
const          785.0985      43.847      17.905      0.000      699.089
871.108
Age              0.4709       1.153       0.408      0.683      -1.790
2.732
```

```
=====
=====
```

```
Omnibus:          1174.453   Durbin-Watson:
2.082
Prob(Omnibus):          0.000   Jarque-Bera (JB):
88.696
Skew:          -0.004   Prob(JB):
5.49e-20
Kurtosis:          1.797   Cond. No.
159.
```

```
=====
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
print("STATISTICAL ANALYSIS SUMMARY")
print("=" * 40)
```

```
print("\n1. T-TEST") print("T-
statistic:", t_stat) print("P-value:",
p_value)
```

```
if p_value < 0.05:
    print("Result: Significant difference in Age.")
else:
    print("Result: No significant difference in
Age.")
```

```
print("\n2. ANOVA") print("F-
statistic:", f_stat) print("P-value:",
p_value_anova)
```

```
if p_value_anova < 0.05:
    print("Result: Significant difference
among Departments.")
else:
    print("Result: No significant
difference among Departments.")
```

```
print("\n3. LINEAR REGRESSION") print("R-
squared:", model.rsquared)
```

```
print("Age coefficient:", model.params["Age"])  
print("Age p-value:", model.pvalues["Age"])
```

STATISTICAL ANALYSIS SUMMARY

=====

1. T-TEST T-statistic: -
5.82801185398895
P-value: 1.3797600649439802e-08
Result: Significant difference in Age.

2. ANOVA

F-statistic: 0.7655031924976907
P-value: 0.4652855299935297
Result: No significant difference among Departments.

3. LINEAR REGRESSION R-squared:
0.00011365569809163034
Age coefficient: 0.47089343162226616
Age p-value: 0.6829722453880929